

the disk alphabet (number) and 'y' is the partition number. Replace 'X' and 'y' as per your drive's values. In my case it was `/dev/sdb1` (X=b, y=1)



Warning: Always be extra careful when executing commands in super-user mode. Double check which device you are operating on when setting boot flags, bootloaders, etc, or you may render your system unbootable.

Phase I: Preparation

Create the `work_dir` directory into which we will dump all the resources. Create `mbcd/` and `boot/` directories in chains inside `work_dir/`. We'll use `mbcd/` to hold the compiled image files, while `boot/` will hold the boot files for the image. Then change directory to `work_dir/` and be sure to remain fixed in this location.

```
$ mkdir -p work_dir/mbcd/boot
$ cd work_dir
```

Download the distros and then copy them into `work_dir/`. Check the *"Links and references"* section at the end of the article for the official websites of the distros.

Create temporary directories in `/mnt/`, which we will use as mount points of the ISO images. (You can mount one image and then unmount it to reuse it, but for easy reference we will mount all the ISOs at once.) Make seven directories, `tmp1` to `tmp7` in `/mnt/`.

```
# mkdir /mnt/tmp{1,2,3,4,5,6,7}
```

We will now mount the ISO images in the mount point directories. Remember to keep track of which OS is mounted in which directory. You can name the directories accordingly, such that you can identify each from its name. But for simplicity, we will refer to them as 'tmpX'.

Austrumi, Puppy Linux, Slitaz, CDlinux, DSL, GeeXboX, and GoblinX (Micro) will be mounted into `tmp1`, `tmp2`, `tmp3`, `tmp4`, `tmp5`, `tmp6` and `tmp7`, respectively. We will set the owner and group of the mounted images to your normal user account and its group, by specifying the UID (user ID) and GID (group ID) options.

As a normal user, run `id -u` and `id -g` to find out the user and group IDs, respectively, and note them down. Replace `<id>` and `<group>` in the following commands with your UID and the GID, respectively.

```
# mount austrumi*.iso /mnt/tmp1 -o loop,uid=<id>,gid=<group>
# mount puppy*.iso /mnt/tmp2 -o loop,uid=<id>,gid=<group>
# mount slitaz*.iso /mnt/tmp3 -o loop,uid=<id>,gid=<group>
# mount CDlinux*.iso /mnt/tmp4 -o loop,uid=<id>,gid=<group>
# mount dsl*.iso /mnt/tmp5 -o loop,uid=<id>,gid=<group>
# mount geexbox*.iso /mnt/tmp6 -o loop,uid=<id>,gid=<group>
```

```
# mount goblinx*.iso /mnt/tmp7 -o loop,uid=<id>,gid=<group>
```

Change the distro ISO file names as per the downloaded file. In this guide the OS versions are: Austrumi 1.7.2, PuppyLinux 4.0, Slitaz 1.0, CDlinux 0.9.1, DSL 4.2.2, GeeXBox 1.2.2, and GoblinX Micro 2.7.



Note: If you decide to mount the images as the root user (without UID, GID options) then do the copy operations as the root. After you have done all the copying, *'chown'* the copied files to the normal user.



Shortcut: You can skip Steps 3 and 4 of this phase, if you plan to use *isomaster* or *file-roller* and extract the required files in the subsequent phases.

Phase II: Copying

Now we will copy the OS files from the mount points to a proper location in our image directory. First let us understand the requirements briefly while copying the files.

Each OS has a kernel and an initial ramdisk file where the kernel is decompressed and gets loaded into the RAM. Then it loads the files needed, mounts the OS filesystem and runs programs to set up the OS.

The kernel and the initial ramdisk files are loaded through the bootloader from the media. There is a filesystem for each OS where all the programs, scripts and configurations are installed. In the mini distros, these are generally compressed, and/or encapsulated in a filesystem. For example, CDlinux and PuppyLinux use squashfs.

There may be more than one such filesystem for an OS, each containing the required files for a certain application software. For example, an additional squashfs module can be downloaded for OpenOffice.org, or Java runtime in PuppyLinux.

Some distros find the compressed filesystems in a certain location relative to the root of the bootable media. For some distros, the locations of the files can be altered, but information about the changed location should be conveyed by passing an appropriate boot parameter when booting, and so some distros can search their files in all the attached disks of the system, automatically.

To know the requirements of a certain OS, you should read the documentation in the official site, discuss it in forums or on the IRC channel. If you understand shell scripts, mount the initial ramdisk of the distro and read the loading script to decipher it.

After copying the OS files, make sure the copied files are accessible to the distros, and the different OS files don't create any conflict. Although we'll relocate some files and directories to avoid conflicts, we will, by and large, try to keep the copying clean with minimum relocations.